Aunsh Bandivadekar

+1 408-982-6580 | Davis, CA, 95616 | aunsh.spb@gmail.com | linkedin.com/in/aunsh | github.com/aunshx | Portfolio

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, R
Frontend: React, Next.js, TanStack, Redux, Tailwind CSS, shadcn/ui, D3.js, Vite, HTML/CSS
Backend: FastAPI, Node.js, Express.js, PostgreSQL/PostGIS, Redis, GraphQL, REST APIs, Python ETL pipelines
AI / Agents: Anthropic SDK, OpenAI API, LangChain, ChromaDB, RAG
Infra & Tooling: GCP, AWS, Azure, Docker, GitHub Actions, CircleCI, Git, Claude Code, Cursor, Copilot
Testing: Pytest, Jest, React Testing Library, Postman (API contract testing)

Experience

University of California, Davis - **Software Developer, Research** **Davis, CA** | **Sep 2024 – Sep 2026**
Full-stack platform work on geospatial and bio-analytics tools used in production

- Led the rebuild of a legacy open-source platform into a multi-route production app, clearing 30% of the ticket backlog
- Redesigned PostGIS query paths with spatial indexing, turning full-table scans into interactive lookups and 3 days into 30 minutes
- Built scheduled ETL running R spatial models on HPC into PostgreSQL, so Node.js APIs served precomputed results
- Shipped a React/FastAPI platform that generates simulations in place of heavy computation for 40 lab researchers
- Replaced manual model configuration with Python and Vue.js tooling, cutting input prep time 90% for the modeling team

mlpal.ai - **Software Engineer (Founding)** **Remote** | **Mar 2024 – Aug 2024**
Early engineer at a pre-seed AI infrastructure startup

- Owned model selection end to end across a FastAPI and Postgres service and its Next.js surface
- Shipped plain-English model search using ChromaDB and OpenAI embeddings, hitting 70% top-1 accuracy on real queries
- Surfaced ranked recommendations using LangChain over the catalog, exposing the cost and context tradeoff behind each pick
- Ran investor demos on the live product, setting product scope with the founders off what those sessions exposed

Prosperix, Inc - **Software Engineer (Frontend)** **Remote** | **Aug 2022 – Aug 2023**
Candidate-facing engagement layer of a hiring marketplace

- Owned the gamification and data-heavy dashboard UI in React and TypeScript, shipping flows that lifted engagement 15%
- Consolidated overlapping GraphQL queries behind a shared cache, cutting redundant network calls on dashboard load
- Delivered the Node.js APIs persisting progress and points state, with Jest suites in CircleCI gating merges

2WheelR - **Technical Analyst (Founder's Office)** **Pune, IN** | **Jul 2021 – Jul 2022**
Product and GTM in the founder's office at a mobility startup

- Established the first product spec from customer interviews rather than founder intuition, at $200K+ ARR
- Tracked weekly conversion and drop-off metrics for the founders, grounding product scoping in real usage

Projects

FRED - Agentic AI for Geospatial Decision Support | *Next.js, FastAPI, LangGraph, XGBoost, PostGIS, GCP* biofred.us

- Built the React/Leaflet frontend streaming partial agent output over SSE with H3 hexbin map overlays
- Cut statewide site screening from an overnight batch run to a live query, standing in learned surrogates for the cost engine
- Designed CS-RAG, composing spatial and economic retrieval operators at runtime so new query types ship without new endpoints
- Rewrote agent control flow as LangGraph state machines checkpointed to Postgres, ending skipped retrievals and lost work
- Shipped the first release solo, now leading a team of 6, with industry partners in evaluation and Canada scoped next

Watchtower - Agentic GTM Pipeline | *Claude Code, Anthropic API, Python, Vite* Github

- Developed a ReAct agent mining 100 OSS repos for AI-coding friction, scoring 2,774 candidates down to 5 qualified leads
- Grounded per-repo outreach in maintainers' own open issues, drafted from agent findings rather than templates
- Instrumented session traces and scheduled nightly runs on GitHub Actions, making every agent decision replayable

ResView - Blockchain Consensus Visualizer | *Open Source* | *Apache ResilientDB, React, D3.js, C++* Github

- Engineered the React/D3.js frontend rendering prepare and commit flows streamed live from the C++ replica layer
- Injected Byzantine faults to break quorum on demand, validating PBFT view changes and recovery in 4-node clusters
- Merged upstream into an Apache Incubator project, now used by 300+ students in UC Davis distributed systems courses

Education

University of California, Davis **Sep 2023 – Jun 2026**
Master of Science in Electrical and Computer Engineering *Davis, CA*
Thesis: Agentic Decision Support via Spatial Prediction Layers

University of Pune **Jun 2016 – May 2020**
Bachelor of Engineering in Electronics and Telecommunication *Pune, India*
Awards: Best Paper – International Conference on Pervasive Computing | INSPIRE Scholarship (top 1%) – Govt. of India